3V8_PRE | TPS62135 forced PWM
1V0_DSP_CORE | TPS62135 forced PWM
3V3_SYS | TPS62135 forced PWM
MODE=VIN; VSEL=0; FB2=0; native PG intentionally NC.
Kelvin FB bottom to GND; VOS to local COUT positive.
L/C qualification and distributed capacitance remain board validation.
1V8_DSP_REF_ANA | TPS7A4901DGNR
1V35_DSP_DMC | TPS7A4901DGNR
5V_AFE | TPS7A4901DGNR
DNC pin 7 and NC pin 3 open; EP pin 9 grounded.
Preload is permanent, included in total rail allocation.
CFF spans OUT to FB (including series top-arm resistors).
3V3_ADC_A | TPS7A2033PDBVR
2V8_MIC | TPS7A2028PDBVR
P variant: active output discharge. Qualified branch EN.
No startup-time guarantee extrapolated to this capacitor bank.
Consumer-side filtering/decoupling belongs to the later load sheets.
Four DSP rail comparators / first-rail qualification
PG1..PG4 = REF (+ADJ); no PG capacitors. RT=VCC: 160..240ms.
RST=RAILS_OK, isolated from SYS_HWRST.
Do not substitute regulator PG; sense returns Kelvin to supervisor GND.
TPS3431: independent watchdog / delayed reset
CWD NC; SET1=VDD; WDO and ENOUT wired open drain.
330..470ms total rail-valid to reset release; WDT 1.36..1.84s.
First DSP falling WDI edge <=1.13s after reset release.
DSP WDI default low; firmware services during boot.
MCLK /OE from RAILS_OK only. Clock bank /1OE from CLOCK_STARTUP_EN.
Data bank /2OE from AUDIO_DATA_OE_RELEASE only.
SN74AXC2T245, LVC244A, ASDLJ/LMK and endpoints: later-sheet contracts.
ADC sheet: DVDD CEXT=10uF X7R, Ceff<=12uF, REXT=3.00k 1%;
PD/RST LOW >=50ms for in-place reset; PWUP only after DVDD+clock >=10ms.
Downstream qualification 1/2
Downstream qualification 2/2
Commanded shutdown clock
Hardware safe clear 1/3
Hardware safe clear 2/3
Hardware safe clear 3/3
Explicit post-check arm edge
Analog qualification 1/2
Analog qualification 2/2
Data authorization; no bootstrap gating
U_SEQ_RUN: retained service-off state
U_SAFE_LATCH: functional authorization
Raw PGOOD / eFuse Schmitt conditioning
Retained shutdown / immediate brownout reset
U_DOWN_DELAY: OV / active-low / open drain
CTS 198..242nF acceptance; hold 20.415..40.946ms.
CTR/MR open. Cold-start reset may add <=2ms.
U_CMD_XLAT: eight isolated A-to-B controls
Amplifier MUTE / STANDBY authorization
U_BOOT_GATES: clock bootstrap / health clear
AFE channel 1 --> ADC, two differential legs
AFE channel 2 --> ADC, two differential legs
AFE channel 3 --> ADC, two differential legs
AFE channel 4 --> ADC, two differential legs
ADC VREF --> AFE VCM buffer; spare channels OFF
Microphone pair 1 --> AFE coupling capacitors
Microphone pair 2 --> AFE coupling capacitors
Used SEL = ANALOG_PWR_EN; either adjacent rail absent isolates path.
No endpoint load circuitry. Keep differential switch paths symmetrical.
START: RAILS_OK --> MCLK --> DSP boot --> CLOCK_STARTUP_CMD --> ADC reset/config/PLL --> RUN_HEALTH_OK + arm edge.
Only after arm: data, amplifier Hi-Z clock check, analog settling, then MUTE/PLAY. ADC-only PLL loss: hardware automute + firmware health withdrawal.
STOP: MUTE/Hi-Z/STANDBY >=15ms --> analog OFF, >=1ms --> data OFF + ADC reset --> health LOW --> clock command LOW --> SHUTDOWN_REQ.
Retained reset --> 20.415..40.946ms --> downstream OFF. Service-off retains pre + 1V8 until external power cycle; no in-place restart.

Phase 3D rails; Phase 3E retained shutdown; Phase 3F bootstrap and functional authorization.
circuitBuilderAi-Astra
Sheet: /
File: power_regulation.kicad_sch
Title: 2 - Power Regulation / Sequencing
Size: User
Date: 2026-09-14
Rev: 4B
KiCad E.D.A. 10.0.6
Id: 1/1